

Battery ECM + EKF Case Study (WLTP)

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Repository: https://github.com/sufiyan-mtu/SOC_Estimation-.git

Models:

- 2-RC ECM with FOH slow branch and FOH $\leftrightarrow$ ZOH fast branch ($|dI/dt|$ hysteresis)
- Scalar voltage-domain SOC EKF (Path-B): R-scheduling with raw OCV slope, NIS gating, trust region, Joseph covariance

Data: data/10-25-19_11.29 960_WLTP206b.mat (WLTP segment)

Dictionary: mdl/ecm_dd.sldd (populated by scripts/setup_ecm.m)

1) Methodology & Intermediate Steps

1.1 Data ingestion & preprocessing

- Load MAT file; extract Time, Current, Voltage.
- Shift time to $t=0$; enforce strictly increasing time; remove duplicates.
- Compute $Ts_data = \text{median}(\text{diff}(t))$.
- Build From-Workspace timeseries: I_A , V_meas .
- Estimate initial SOC by inverting OCV (percent $\rightarrow$ fraction) to get $SoC0_guess$.
- Conventions: current is discharge-positive.

1.2 ECM parameterization & LUT build (scripts/setup_ecm.m)

- Read data/ecm_lookup_table.csv with columns: SOC, C_rate, R0, R1, C1, R2, C2.
- Build scattered-interpolant surfaces over (SOC, C-rate); evaluate on dense grids.
- Write to DD: soc_bp, crate_bp, R0_lut, R1_lut, C1_lut, R2_lut, C2_lut, OCV_SOC_bp (%), OCV_lut (V).

2) ECM Discretization & Equations

Terminal equation (discharge-positive):

$$V_ter(k) = OCV(SOC_k) - R0 I_k - V_rc1,k - V_rc2,k$$

ZOH update (exact for staircase input):

- $a = \exp(-Ts/\tau)$, $\tau = R \cdot C$
- $V_k = a V_{k-1} + R(1-a) I_{k-1}$

FOH update (exact for linear ramp within the step):

- $a = \exp(-T_s/\tau)$
- $B_{\text{prev}} = R - R*(\tau/T_s)*(1-a)$
- $B_{\text{cur}} = R*(\tau/T_s)*(1-a) - R*a$
- $V_k = a V_{\{k-1\}} + B_{\text{prev}} I_{\{k-1\}} + B_{\text{cur}} I_k$
- Constraint: $B_{\text{prev}} + B_{\text{cur}} = R(1-a)$

Fast-branch mode selection (FOH↔ZOH with hysteresis):

- If last mode = ZOH and $|dI/dt| > \theta \rightarrow$ switch to FOH.
- If last mode = FOH and $|dI/dt| < 0.7 \theta \rightarrow$ switch to ZOH.
- Else hold the previous mode. Slow branch uses FOH only.

3) EKF with Equations

State, prediction, and measurement:

- $x_k|k-1 = x_{\{k-1\}} - (\eta T_s/(Q_n 3600)) I_k$
- $P_k|k-1 = P_{\{k-1\}} + Q_{\text{proc}}$

R-scheduling (uses raw slope for plateaus):

$$R_{\text{eff}} = R_{\text{meas}} (1 + \alpha_R |I_k|) (1 + \alpha_{\text{plateau}} (H_{\text{ref}}/|H_{\text{raw}}|)^2)$$

Innovation, gating, update (trust-region), covariance:

- $v_k = z_{\text{meas}} - z_k$, $S_k = H^2 P_k|k-1 + R_{\text{eff}}$, $K_k = (P_k|k-1 H)/S_k$
- If $NIS = v_k^2 / S_k > 2.71 \rightarrow$ set $K_k = 0$ (gate)
- $\Delta x = \text{clip}(K_k v_k, \pm 0.005)$, $x_k = \text{sat}_{[0,1]}(x_k|k-1 + \Delta x)$
- $P_k = (1 - K_k H)^2 P_k|k-1 + K_k^2 R_{\text{eff}}$ (Joseph form)

De-polarized measurement fed to EKF:

$$V_{\text{OCV}}^{\text{meas}} = V_{\text{ter}} + V_{\text{rc1}} + V_{\text{rc2}} + R_0 I$$

4) Methodology Assumptions

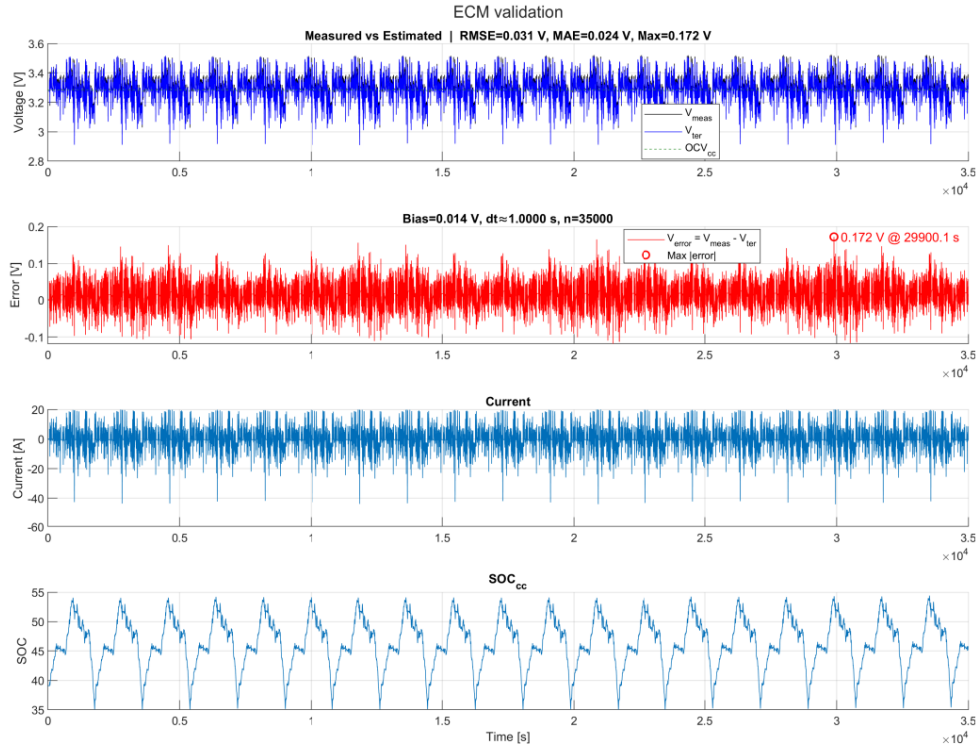
- Single-population ECM: R/C and OCV are temperature-agnostic and SoH-agnostic.
- Dataset cycles three different cells; no cell-to-cell variability, aging/SoH, or temperature dependence modeled in ECM.
- Implication: measurement path may be biased; EKF uses larger base R_{meas} (0.36 V^2) and raw-slope R_{eff} inflation to prefer prediction on plateaus.

5) Decision Justification

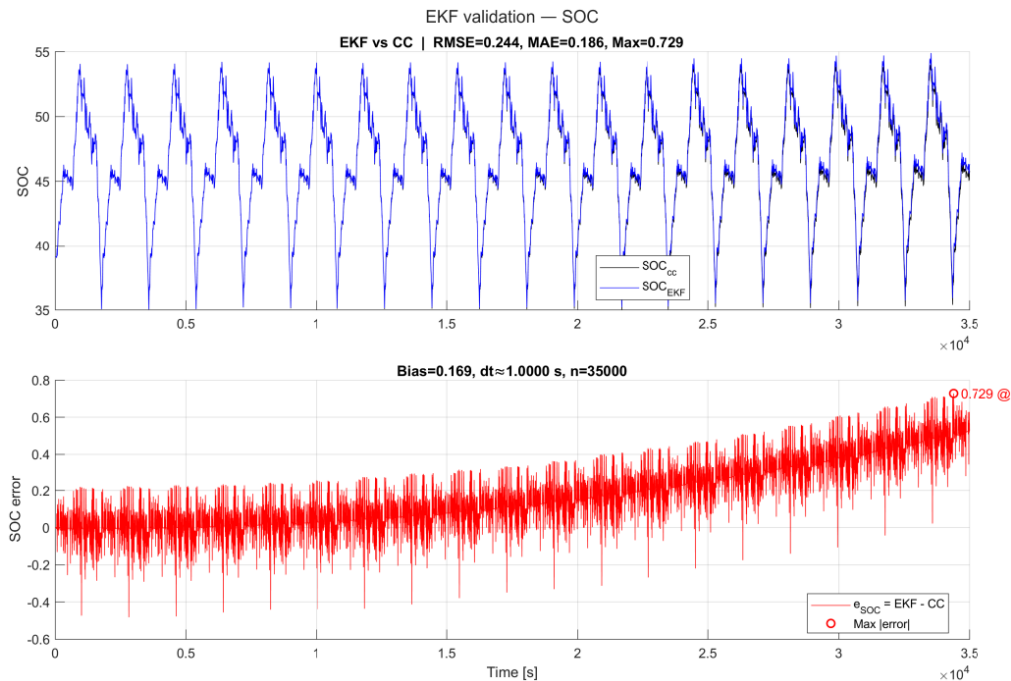
- FOH on slow; FOH↔ZOH on fast to cover ramps and steps.
- Raw-slope R-scheduling inflates R_{eff} on plateaus; floored H stabilizes the update.
- Larger $R_{\text{meas}} = 0.36 \text{ V}^2$ for robustness to unmodeled variability.
- NIS gating + trust region for bounded, outlier-resistant updates.
- FOH coefficients kept simple for clarity; adequate accuracy for tested T_s/τ .

6) Results & Key Observations

- Simulation length: duration 34999.16 s, $dt \approx 1.0000$ s, $n = 35000$.
- ECM: RMSE 0.030662 V, MAE 0.023560 V, Max |err| 0.172098 V @ 29900.139 s, Bias 0.013723 V.

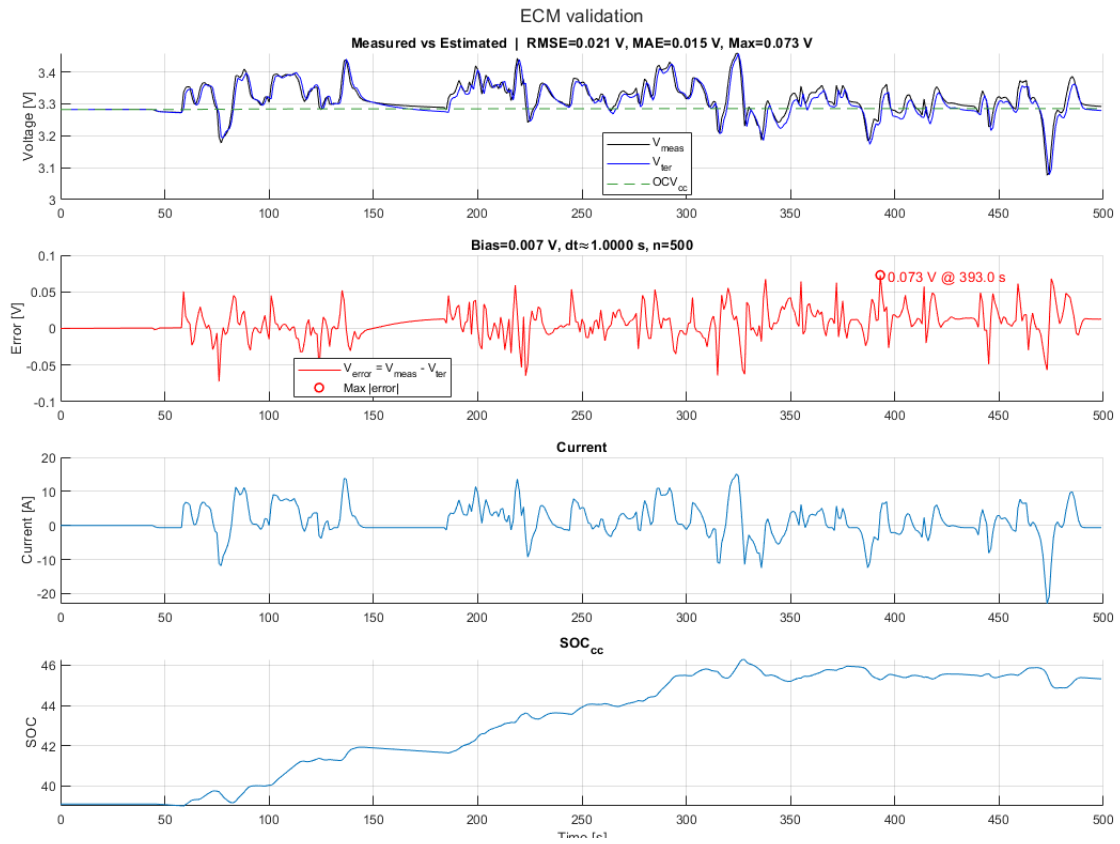


- EKF: SOC RMSE 0.243860, MAE 0.185798, Max |err| 0.728614 @ 34385.2 s, Bias 0.168538;

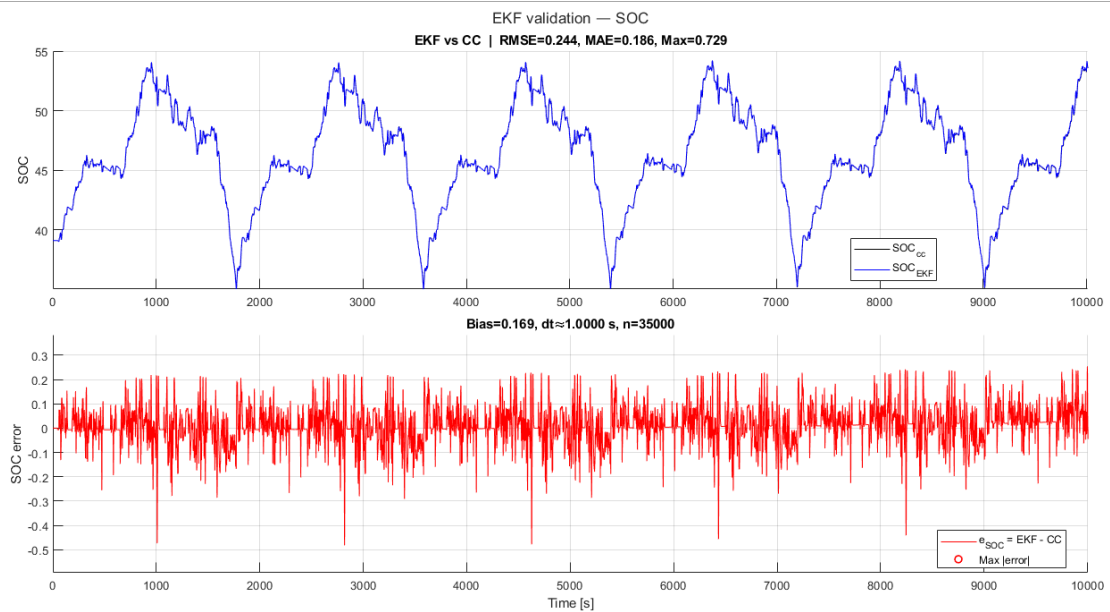


7) Plot Guide

- ECM Measured vs Simulated Voltage: ECM fidelity; residuals correlate with load.



- EKF Estimated vs Reference SOC: stable SOC with bounded corrections.



8) Discussion

- Worked: FOH↔ZOH fast branch; raw-slope R_{eff} ; NIS gating + trust region → smooth, bounded SOC.
- Trade-off: single, temperature/SoH-agnostic ECM with three-cell dataset → larger R_{meas} ; slower convergence but more robust.
- Limitations: no temperature/aging modeling

9) Conclusion

- The 2-RC ECM (FOH on slow, FOH↔ZOH on fast) plus Path-B EKF with raw-slope scheduling and conservative R_{meas} achieves low voltage error and stable, bounded SOC on WLTP.
- Predictive weighting on plateaus and gradual corrections on informative slopes match the dataset and modeling assumptions (temperature/SoH ignored).